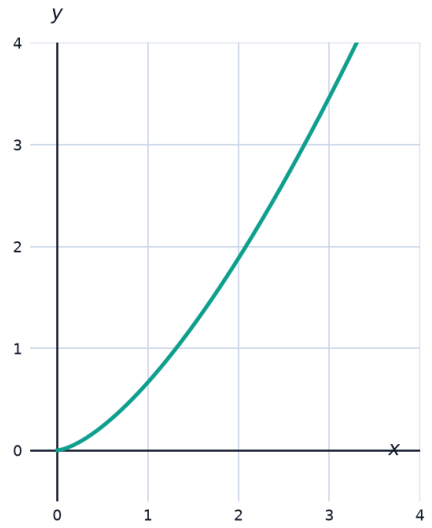
Arc Length and Improper Integrals

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.03

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

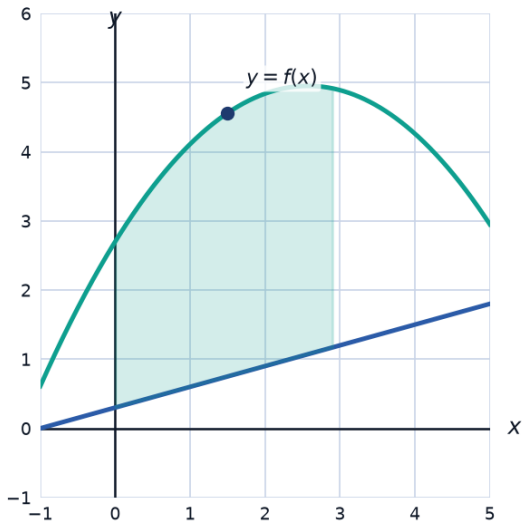
1 [Foundational] • Target: Arc length

Find the arc length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.



2 [Foundational] • Target: Arc length

Set up (do not evaluate) the arc length integral for $y = \sin x$ on $[0, \pi]$.



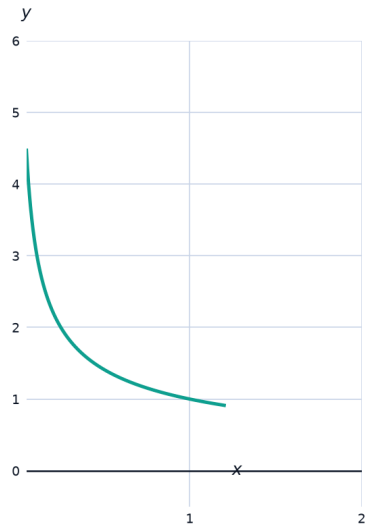
3 [Foundational] • Target: Improper integrals

Evaluate $\int_1^\infty \frac{1}{x^2}\,dx$, or state that it diverges.

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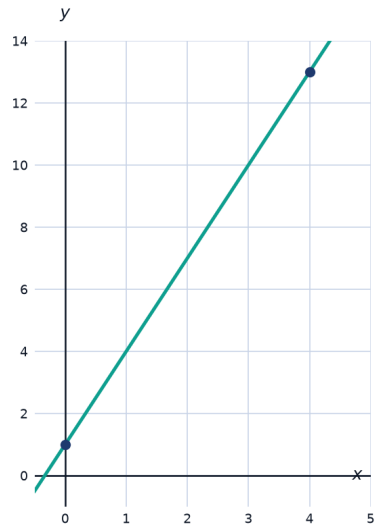
4 [Foundational] • Target: Improper integrals

Evaluate $\int_0^1 \frac{1}{\sqrt{x}}\,dx$ (improper at the lower limit).



5 [Foundational] • Target: More arc length practice

Find the arc length of the line $y = 3x + 1$ from $x = 0$ to $x = 4$, and verify your answer using the distance formula.



6 [Foundational] • Target: The p-integral test

For $\int_1^\infty \frac{1}{x^p}\,dx$, state the condition on p for convergence, and explain why using the antiderivative $\frac{x^{1-p}}{1-p}$ (for $p \neq 1$).

7 [Foundational] • Target: The p-integral test

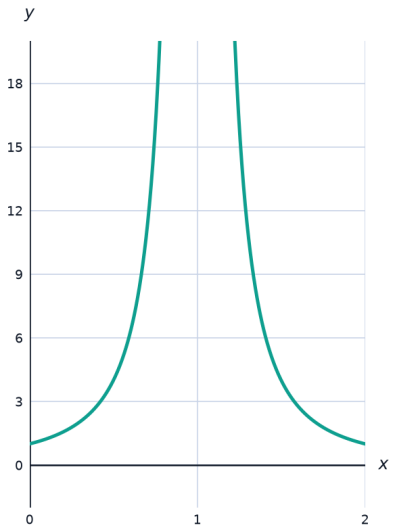
Use the p-integral result to quickly determine whether $\int_1^\infty \frac{1}{x^{0.9}}\,dx$ converges.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Improper integrals with infinite discontinuities inside the interval

Evaluate $\int_0^2 \frac{1}{(x-1)^2}\,dx$, or explain why it diverges (note the integrand is undefined at $x = 1$, inside the interval).

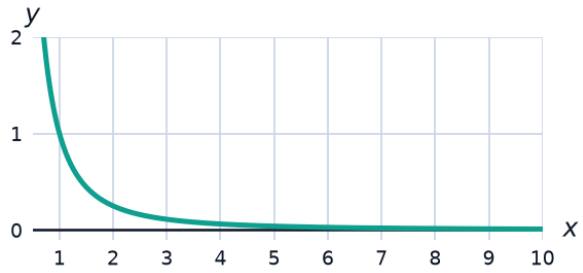


9 [Intermediate] • Target: Improper integrals with infinite discontinuities inside the interval

Evaluate $\int_{-1}^{1} \frac{1}{x^{1/3}}\,dx$ (an odd function with a discontinuity at $x = 0$), splitting at $x = 0$.

10 [Intermediate] • Target: Visualizing an improper integral as a limiting area

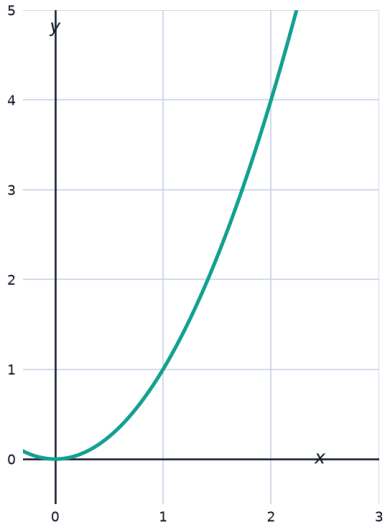
The graph shows $f(x) = \frac{1}{x^2}$ on $[1, 10]$. As the right boundary extends to infinity, the shaded area approaches a finite limit. Explain why this is plausible, using the fact that $\int_1^{\infty} \frac{1}{x^2}\,dx = 1$ (found earlier).



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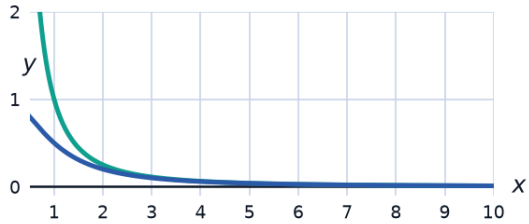
11 **[Intermediate]** • **Target: Arc length of a more involved curve**

Set up (do not evaluate) the arc length integral for $y = x^2$ on $[0, 2]$, and estimate it using the straight-line distance between the endpoints as a lower bound.



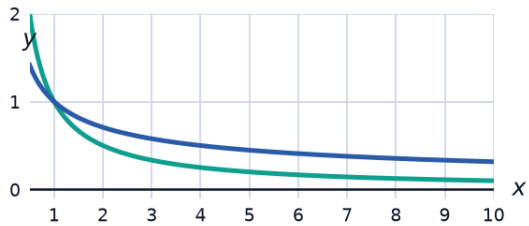
12 **[Intermediate]** • **Target: Comparison test for improper integrals**

Use a comparison with $\int_1^\infty \frac{1}{x^2}\,dx$ (known to converge) to argue that $\int_1^\infty \frac{1}{x^2+1}\,dx$ also converges, without evaluating it.



13 **[Intermediate]** • **Target: Comparison test for improper integrals**

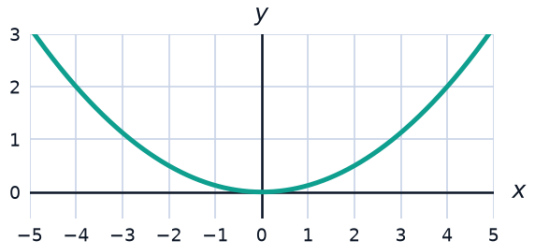
Use a comparison with $\int_1^\infty \frac{1}{x}\,dx$ (known to diverge) to argue that $\int_1^\infty \frac{1}{\sqrt{x}}\,dx$ also diverges.



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14 [Intermediate] • Target: A capstone arc-length application

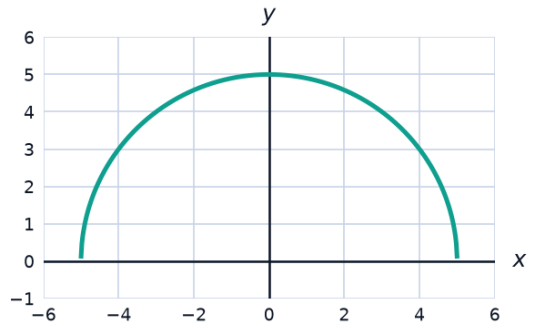
A cable hangs in the shape $y = \frac{x^2}{8}$ between the towers at $x = -4$ and $x = 4$. Set up the integral for the length of the cable (do not evaluate).



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: A capstone arc-length application • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

A city's water tower is shaped by revolving the curve $y = \sqrt{25 - x^2}$ (the upper half of a circle of radius 5) about the x-axis; set up (do not evaluate) the arc length integral for this curve on $[-4, 4]$, which describes the profile of one side of the tower.



16 [Challenging] • Target: Improper integrals with infinite discontinuities inside the interval • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Level 2 · MediumApply the Ratio Test to determine the convergence or divergence of $\sum_{n=1}^{\infty} \frac{4^n}{n!}$.

17 [Challenging] • Target: Visualizing an improper integral as a limiting area • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

[Unit 9 · Polar Coordinates & Polar Area · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

18 [Challenging] • Target: Arc length of a more involved curve • Cross-Lesson: 6.01 Related Rates

[Unit 10 · Infinite Sequences & Series — Convergence Tests · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

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19 [Challenging] • Target: Comparison test for improper integrals • Cross-Lesson: 6.02 Optimization and Curve Sketching

Level 1 · FoundationsFind all values of c in the open interval $(-1, 2)$ that satisfy the Mean Value Theorem for $f(x) = x^3 - 2x^2 + 1$.

20 [Challenging] • Target: A capstone arc-length application • Cross-Lesson: 7.01 Integration by Substitution

A random variable X has distribution $\text{Po}(\lambda)$. A single observation of X is taken to test $H_0 : \lambda = 6$ against $H_1 : \lambda < 6$ at the 5% level. (a) Find the critical region. (b) State the actual significance level. (c) Find the probability of a Type II error if the true value of λ is 3.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Arc length

Comparison test for improper integrals: $0 \le f(x) \le g(x)$, $\int_a^\infty g$ converges implies:

a. $\int_a^\infty f$ converges

b. $\int_a^\infty f$ diverges

c. $f = g$

d. nothing

22 [Foundational] • Target: Improper integrals

If $f'(x) = (x - 3)$, on which interval is f increasing?

a. $(3, \infty)$

b. \{3\}

c. $(-\infty, \infty)$

d. $(-\infty, 3)$

23 [Foundational] • Target: More arc length practice

Polar area inside $r = 1 + \cos\theta$ and outside $r = 1$:

a. $2 + \pi/4$

b. $\pi/4 + 1$

c. $1 + \pi/4$

d. $\pi/2$

24 [Foundational] • Target: More improper integrals

Polar: find the area inside $r = 3\sin\theta$ and outside $r = 1 + \sin\theta$:

a. $\pi/4$

b. π

c. $\pi/2$

d. $3\pi/4$

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25 [Foundational] • Target: The p-integral test

A box with square base and open top must hold 32 m³. Minimise surface area. Optimal side length x:

a. 2

b. 4

c. $2\sqrt[3]{4}$

d. $\sqrt[3]{32}$

26 [Foundational] • Target: Improper integrals with infinite discontinuities inside the interval

Area inside $r = 2\cos\theta$ and outside $r = 1$:

a. $\frac{\pi}{2} + \frac{\sqrt{3}}{2}$

b. $\frac{\pi}{3} + \frac{\sqrt{3}}{2}$

c. $\pi/3$

d. $\pi/2$

27 [Foundational] • Target: Visualizing an improper integral as a limiting area

If $f'(x) > 0$ on an interval, f is:

a. concave down

b. decreasing

c. increasing

d. constant

28 [Foundational] • Target: Arc length of a more involved curve

If $f''(x) > 0$ on an interval, the graph of f is:

a. concave up

b. decreasing

c. concave down

d. linear

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Comparison test for improper integrals

Arc length of $r = 2\cos\theta$ (full circle, $0 \le \theta \le \pi$):

a. π

b. 4

c. $\pi/2$

d. 2π

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30 [Intermediate] • Target: A capstone arc-length application

For the catenary $y = a\cosh(x/a)$, the arc length from $-L$ to L is:

a. $2L$

b. $2a\sinh(L/a)$

c. $L\cosh(L/a)$

d. $2a\cosh(L/a)$

31 [Intermediate] • Target: Arc length

Area swept inside one loop of $r^2 = 4\sin 2\theta$:

a. 1

b. 2

c. 4

d. π

32 [Intermediate] • Target: Improper integrals

Comparison test: if $0 \le a_n \le b_n$ and $\sum b_n$ converges, then $\sum a_n$:

a. diverges

b. oscillates

c. converges

d. is zero

33 [Intermediate] • Target: More arc length practice

$\int_0^{\pi}\int_0^{\sin\theta} r\,dr\,d\theta$: inner integral gives $\sin^2\theta/2$; outer gives

a. π

b. $\pi/2$

c. $\pi/4$

d. $1/2$

34 [Intermediate] • Target: More improper integrals

Triple integral in cylindrical: $\int_0^{2\pi}\int_0^1\int_0^1 r\,dz\,dr\,d\theta =$

a. 4π

b. 2π

c. $\pi/2$

d. π

35 [Intermediate] • Target: The p-integral test

Arc length of helix $\mathbf{r}(t) = (\cos t, \sin t, t)$, $0 \le t \le 2\pi$:

a. 2π

b. $\sqrt{2}$

c. 4π

d. $2\pi\sqrt{2}$

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36 [Intermediate] • Target: Improper integrals with infinite discontinuities inside the interval

Arc length parametric: $L = \int_a^b \sqrt{\dot{x}^2 + \dot{y}^2}\, dt$. For $x = t$, $y = t$ from 0 to 1:

a. $\sqrt{2}$

b. 1

c. 2

d. $\sqrt{2}/2$

37 [Intermediate] • Target: Visualizing an improper integral as a limiting area

Probability: $f(x, y) = 6xy$ on $0 \le x \le 1, 0 \le y \le \sqrt{1 - x^2}$ (quarter circle). $P(X + Y \le 1) =$

a. $\frac{3}{4}$

b. $\frac{1}{2}$

c. $\frac{1}{4}$

d. $\frac{2}{3}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Arc length of a more involved curve • Cross-Lesson: 6.01 Related Rates

Use $x = u + v$, $y = u - v$ to compute $\iint_D (x - y)e^{x+y}\, dA$ where D is the square $|x - y| \le 1, |x + y| \le 1$:

a. 0

b. $2(e - e^{-1})$

c. $(e - 1)^2$

d. $4(e - e^{-1})/e$

39 [Challenging] • Target: Comparison test for improper integrals • Cross-Lesson: 6.02 Optimization and Curve Sketching

A ladder 10 m long leans against a wall; the base slides away at 1 m/s. How fast is the top sliding down when the base is 6 m from the wall?

a. $\frac{3}{4}$

b. $-\frac{3}{8}$

c. $-\frac{3}{4}$

d. $-\frac{3}{2}$

40 [Challenging] • Target: A capstone arc-length application • Cross-Lesson: 7.01 Integration by Substitution

Evaluate $\iiint_B (x^2 + y^2 + z^2)\, dV$ where B is the ball $x^2 + y^2 + z^2 \le 1$.

a. $\frac{4\pi}{5}$

b. $\frac{4\pi}{3}$

c. $\frac{2\pi}{5}$

d. $\frac{8\pi}{5}$

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41 [Challenging] • Target: Arc length • Cross-Lesson: 7.02 Integration by Parts

Variation of parameters for $y'' + y = \sec x$: general solution involves $\ln|\sec x + \tan x|$. This appears because the particular integral of $\int \sin x \sec x \, dx =$

a. $\tan x$

b. $\ln|\cos x|$

c. $-\ln|\cos x|$

d. $-\cos x$

42 [Challenging] • Target: Improper integrals • Cross-Lesson: 8.01 Area Between Curves

The Jacobian of the transformation $x = r\cosh u, y = r\sinh u$ is r times:

a. $\cosh^2 u - \sinh^2 u = 1$

b. $\cosh u + \sinh u$

c. $\sinh^2 u$

d. r^2

43 [Challenging] • Target: More arc length practice • Cross-Lesson: 8.02 Volumes of Revolution

Surface integral $\iint_S c \, dS$ for constant c equals:

a. $c \cdot \text{Area}(S)$

b. c

c. 0

d. $c \cdot \text{Vol}$

44 [Challenging] • Target: More improper integrals • Cross-Lesson: 9.01 Separable Differential Equations

Centre of mass of lamina $D = [0,1] \times [0,1]$ with density $\delta = xy$:

a. $(1/2, 1/2)$

b. $(2/3, 2/3)$

c. $(3/4, 3/4)$

d. $(1, 1)$

45 [Challenging] • Target: The p-integral test • Cross-Lesson: 1.01 Precalculus Review for Calculus

Use polar coordinates to evaluate $\iint_R (x^2 + y^2)^3 \, dA$, where R is the disk $x^2 + y^2 \le 9$.

a. $\frac{6561}{4}$

b. $\frac{6561\pi}{4}$

c. $\frac{6561\pi}{2}$

d. $\frac{6561\pi}{8}$

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